

TASK 2: Array Manipulation.

Given an array $arr[]$, find the prefix sum of the array. A prefix sum array is another array $prefixSum[]$ of the same size such that $prefixSum[i]$ is $arr[0] + arr[1] + arr[2] + \dots + arr[i]$

Sample Input - 1: 5

10 20 10 5 15

Sample Output - 1:

10 30 40 45 60

Explanation: for each index i , add all the elements from 0 to i .

Aim: To design and implement a Java program that takes an array of integers as input and computes its Prefix Sum Array where:

$$prefixSum[i] = \sum_{j=0}^i arr[j]$$

Algorithm:

- (1) Start
- (2) Input the size of the array n .
- (3) Create an array arr of size n and take n integers as input.
- (4) Create a new array $prefixSum$ of the same size
- (5) Set the first element: $prefixSum[0] = arr[0]$.
- (6) Loop through the array from index $i=1$ to n .
 * Calculate $prefixSum[i] = prefixSum[i-1] + arr[i]$.
- (7) Print the elements of the prefix sum Array
- (8) End.

Program:

```
import java.util.Scanner;
public class PrefixSum {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array:");
        int n = sc.nextInt();
        int[] arr = new int[n];
        int[] prefixSum = new int[n];
        System.out.println("Enter array elements:");
        for (int i=0; i<n; i++) {
            arr[i] = sc.nextInt();
        }
        prefixSum[0] = arr[0];
        for (int i=1; i<n; i++) {
            prefixSum[i] = prefixSum[i-1] + arr[i];
        }
    }
}
```

Output

Enter the size of the array : 5

Enter the array elements :

10 20 10 5 15

Prefix Sum Array :

10 30 40 45 60

```

system.out.println("Prefix Sum Array:");
for (int x : prefixSum) {
    System.out.print(x + " ");
}
se.close();
}
}

```

VELTECH	
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Result: Thus, the program is executed ~~and~~ successfully.